

Section A: Conceptual Understanding

- **What is Django REST Framework?**

Django REST Framework (DRF) is used to create APIs in Django.

API means:

- Frontend apps
- Mobile apps
- React apps
- Flutter apps

can communicate with backend/database using JSON.

Example:

```
{  
  "name":"Rahul",  
  "specialization":"Cardiologist",  
  "city":"Surat"  
}
```

1. Why REST APIs are preferred for Mobile & Frontend Apps?

REST APIs are lightweight and use JSON.

Benefits:

- Fast communication
- Works with Android, React, Flutter
- Frontend and backend separated
- Easy integration
- Scalable

Example:

- React frontend
- Django backend
- Communication through API

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2. Serializers as Validation Layer

Serializer converts:

- Python object → JSON
- JSON → Python object

It also validates data.

Example:

```
class DoctorSerializer(serializers.ModelSerializer):  
    class Meta:  
        model = Doctor  
        fields = '__all__'
```

Custom Field Validation

Example:

Doctor name should contain only letters.

```
def validate_name(self, value):  
    if not value.isalpha():  
        raise serializers.ValidationError(  
            "Name should contain only alphabets"  
        )  
    return value
```

3. Importance of HTTP Status Codes

Status codes tell frontend what happened.

Code	Meaning
200 OK	Data fetched successfully
201 Created	New data created
400 Bad Request	Invalid data
404 Not Found	Data not found
500 Server Error	Backend issue

Example:

POST API should return:

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status=status.HTTP_201_CREATED

because new doctor record is created.

4. Why Pagination is Required?

Suppose database has:

- 50,000 doctors

Without pagination:

- API becomes slow
- Huge data load
- High memory usage

Pagination divides data into pages.

Example:

- Page 1 → 10 doctors
- Page 2 → next 10 doctors

Benefits:

- Better performance
- Faster response
- Reduced database load

5. Benefits of ViewSets

Without ViewSet:

- Separate API for GET
- Separate API for POST
- Separate API for DELETE

Too much code.

With ModelViewSet:

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- All CRUD operations automatically generated.

Example:

```
class DoctorViewSet(viewsets.ModelViewSet):  
    queryset = Doctor.objects.all()  
    serializer_class = DoctorSerializer
```

Very fast development.

6. transaction.atomic()

Used for database safety.

If one operation fails:

- entire transaction rollback

No partial data saved.

Example:

```
from django.db import transaction
```

```
with transaction.atomic():  
    serializer.save()
```

Benefits:

- Data integrity
- Prevent corrupted data
- Production safe